TAMIL NADU STATE BOARD - STANDARD 9 SCIENCE (CHEMISTRY)

CHAPTER 1: MATTER AND ITS PROPERTIES

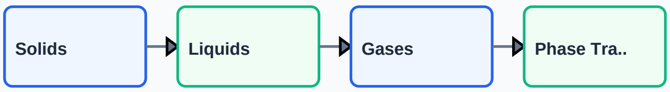
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 9 English Medium
Subject:	Science (Chemistry)
Chapter Name:	Matter and Its Properties

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

TOPIC ROADMAP: STATES OF MATTER & CHARACTERISTICS



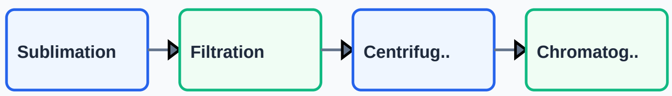
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Matter	Anything that has mass and occupies space, existing in three physical states: solid, liquid, and gas.
Sublimation	The transition of a substance directly from the solid to the gas state without passing through the liquid state.
Centrifugation	A separation technique that rotates mixtures at high speed to separate suspended solids of varying density from liquids.
Chromatography	A laboratory technique for separating mixture components based on differences in adsorption and solubility in a mobile phase.
Homogeneous Mixture	A mixture that has a uniform composition throughout its mass, displaying single-phase boundaries (e.g. air, alloy).

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

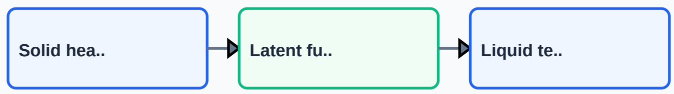
MATTER PHASES SEPARATION FLOW



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

FORMULA: LATENT HEAT OF FUSION TRANSITION



Essential Formulas, Laws & Identities:

- Chromatography Retention Factor (Rf) = Distance traveled by component / Distance traveled by solvent front
- Mass Percentage of solution = (Mass of solute / Mass of solution) * 100
- Mass-Volume Percentage = (Mass of solute / Volume of solution) * 100

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: A solution contains 40 g of common salt in 320 g of water. Calculate the concentration in terms of mass percentage.

Step-by-step Solution:

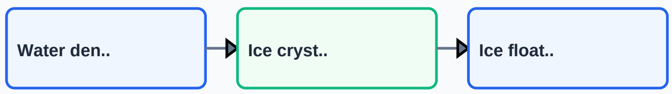
Mass of solute = 40 g, Mass of solvent = 320 g. Mass of solution = Mass of solute + Mass of solvent = 40 + 320 = 360 g. Mass % = (40 / 360) * 100 = 11.11%.

Example 2: In a paper chromatography test, a blue dye traveled 4 cm while the solvent front traveled 10 cm. Find the Rf value.

Step-by-step Solution:

Rf = Distance by dye / Distance by solvent = 4 cm / 10 cm = 0.4.

WORKED: EXPLAINING ANOMALY OF ICE DENSITY



HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Why is water considered a compound while air is classified as a mixture? Explain.

Analytical Solution Strategy:
Water is composed of hydrogen and oxygen chemically combined in a fixed ratio (1:8 by mass), and its properties are entirely different from hydrogen and oxygen. Air consists of nitrogen, oxygen, and other gases physically mixed in variable ratios, and each gas retains its individual physical properties.

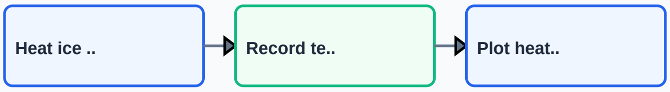
HOTS: BOILING POINT VARIATIONS UNDER PRESSURE CHANGES

Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

- **Q1 (1-Mark):** Compare the properties of solids, liquids, and gases in terms of shape, volume, and compressibility.
- **Q2 (2-Mark):** Explain the separation of ammonium chloride and sand by sublimation.
- **Q3 (2-Mark):** Detail the principle of fractional distillation and name one industrial application.
- **Q4 (5-Mark):** Classify these as homogeneous or heterogeneous mixtures: milk, salt solution, brass, mud water.
- **Q5 (5-Mark):** How can we separate a mixture of oil and water? Describe the apparatus.

PRACTICE: PHASE CHANGE TEMPERATURE LOG SHEET



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Petroleum Refineries

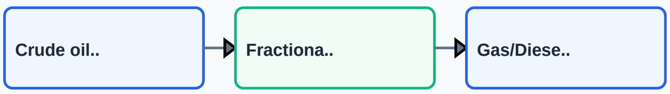
Fractional distillation columns separate crude oil into gasoline, kerosene, diesel, and asphalt.

Application Case: Forensic Laboratories

Paper and thin-layer chromatography are used to separate ink pigments or analyze blood toxins.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

APPLICATION: FRACTIONAL DISTILLATION IN PETROLEUM REFINING



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.

✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.

✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3, N) for the final numeric answer.

✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

[ ] Read and memorized core definitions on Page 2.

[ ] Reviewed critical formulas and relations on Page 3.

[ ] Solved worked examples on Page 4.

[ ] Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress. Good luck!

REVISION: STATES OF MATTER TRANSITIONS